

Acute Neuromuscular Electrical Stimulation Enhances EMG Median Frequency and Delays Fatigue in Individuals with Rotator Cuff Disease

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Introduction

- Individuals with rotator cuff disease (RCD) often experience early onset of fatigue during sustained or repetitive shoulder activities.
- Electrical stimulation evokes involuntary muscle contractions. Beyond strengthening effects, it can also enhance corticospinal excitability and change motor unit recruitment patterns [1], [2], [3].

Research Goal

Our goal is to determine whether a single session of pulsed directed current (PDC) applied to the supraspinatus (SS) and infraspinatus (IS) could attenuate EMG median frequency (MDF) decline during a sustained isometric shoulder abduction task in individuals with RCD.

Results

- No difference in baseline MVC and MDF of both SS and IS between PDC and sham sessions.
- Endurance time (ET) was significantly longer after PDC compared to sham:
PDC: 224.4 ± 19.7 s, Sham: 193.8 ± 17.3 s, $p < 0.05$.
- MF decline (Δ MF) was smaller in PDC than in sham, indicating less fatigue:
SS: -7.7 ± 4.2 Hz (PDC) vs. -10.9 ± 5.1 Hz (Sham), $p < 0.01$
IS: -5.1 ± 3.9 Hz (PDC) vs. -9.7 ± 4.8 Hz (Sham), $p < 0.05$
- The rate of MF decline (slope) was significantly slower with PDC in both muscles.

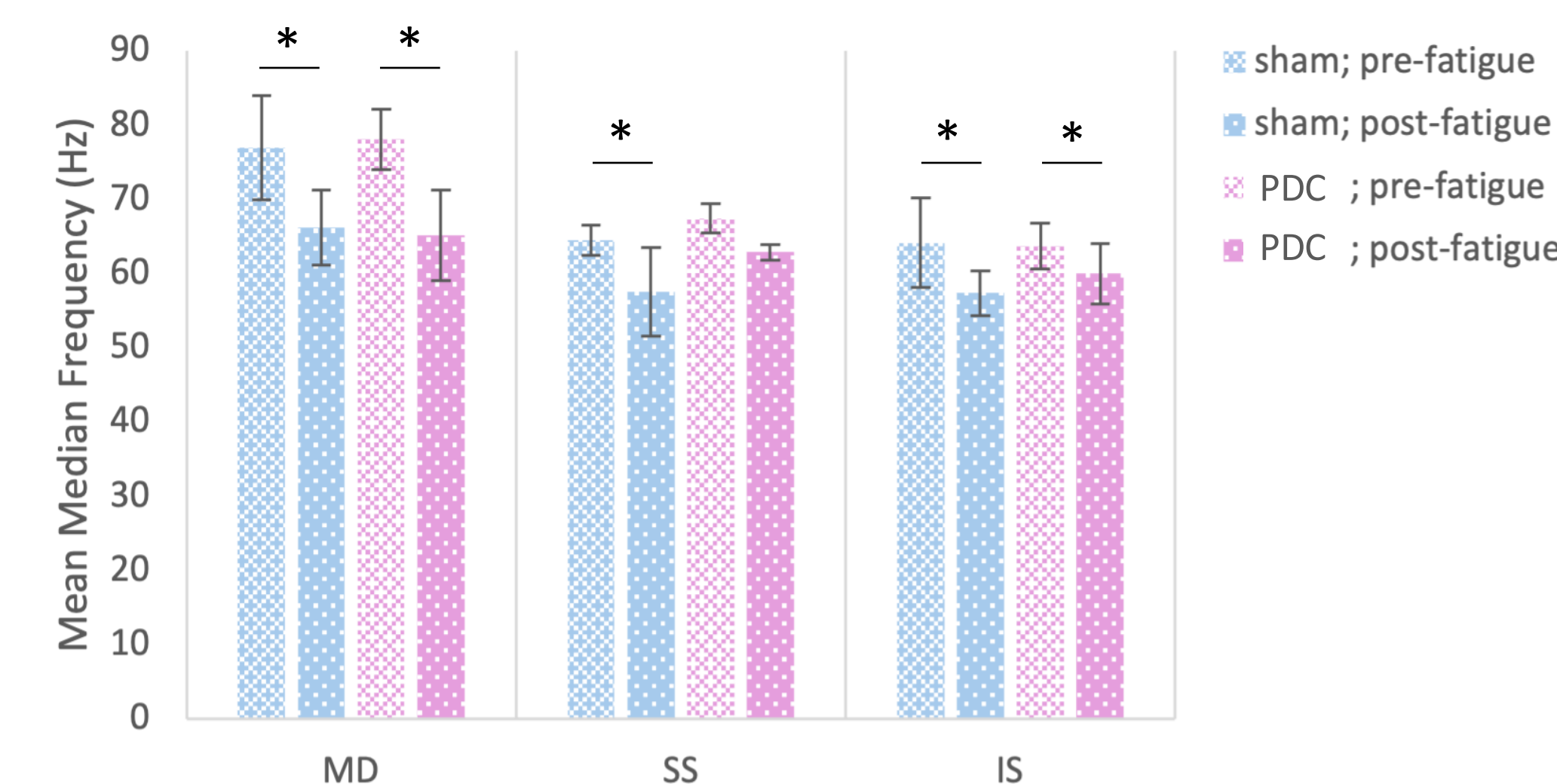


Figure 6. Mean EMG MDF of SS and IS in the first 30s (pre-fatigue) and last 30s (post-fatigue) of contraction after receiving treatment conditions.

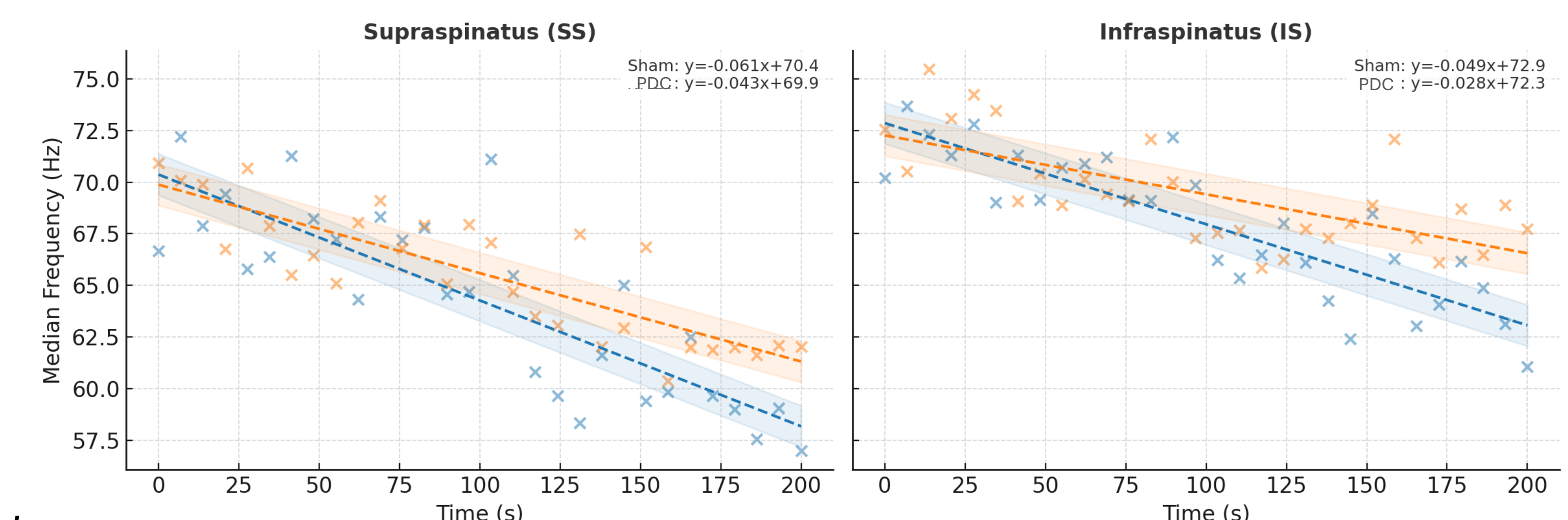


Figure 7. Median frequency declines with time from 0s to 200s.

Methods

- 19 participants (8 male; 28 ± 6 yrs) with RCD were recruited.
- Each participant received a 20-minute treatment on each testing day:
 - Sham session:** 20 min sham stimulation (0 mA)
 - PDC session:** 20 min active PDC (40 Hz, 0.46 ms, maximal tolerated intensity)
- Stimulation was applied to the SS and IS using a NEUBIE stimulator (NeuFit, Austin, TX).
- After treatment, participants performed a sustained isometric shoulder abduction at 25% MVC, with intramuscular EMG collected from SS and sEMG collected from IS and middle deltoid (MD), until task failure.
- EMG MDF was computed every 1s throughout the fatigue task. The rate of MDF decline over time (slope) was used as the **fatigue index**. Independent t-tests assessed whether MDF slopes differed from zero. Two-way repeated-measures ANOVA (Treatment $\times$ Muscle) compared fatigue indices across muscles and between groups.



Figure 2. Neubie Stimulator.

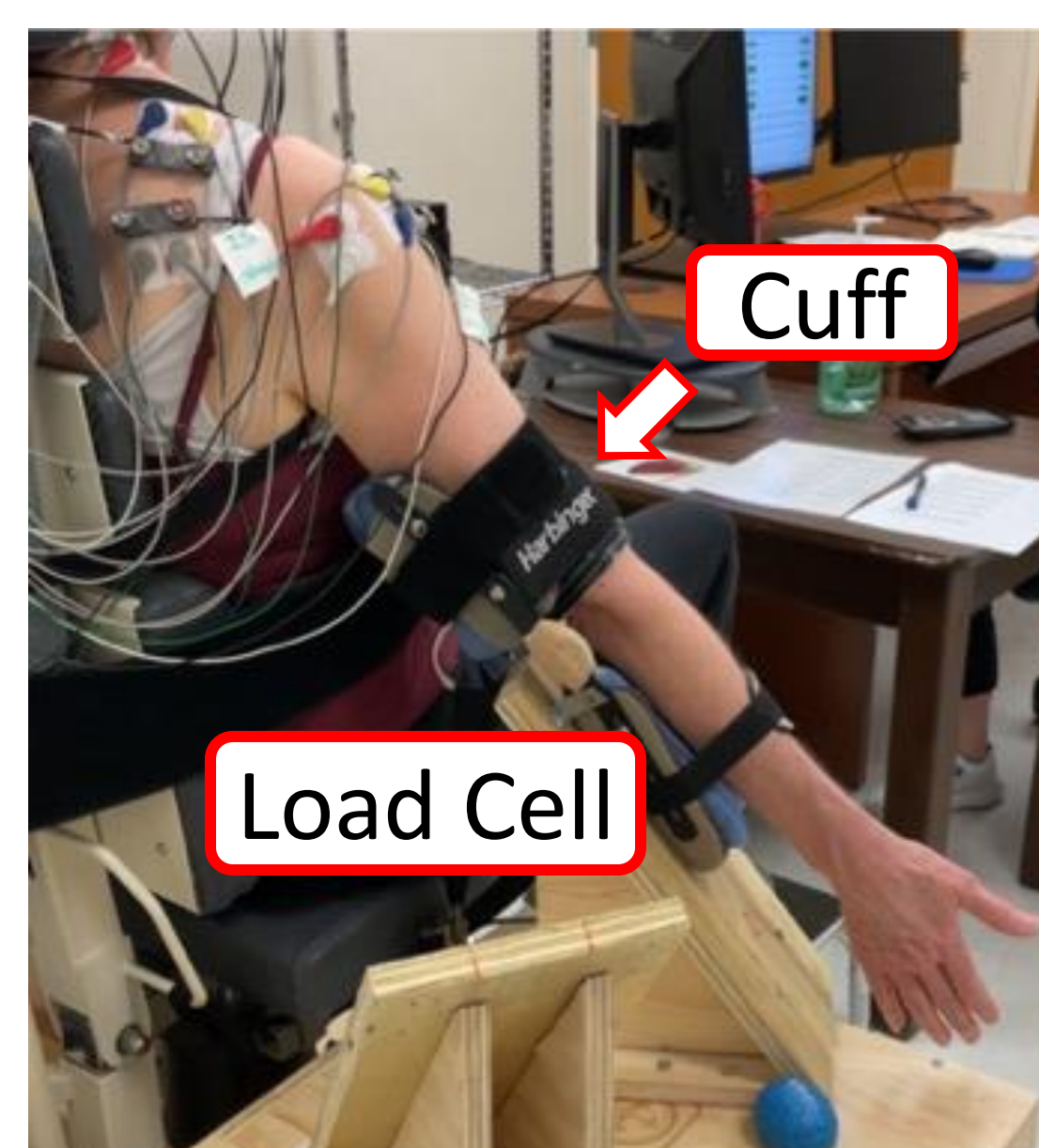


Figure 3. Experimental setting.



Figure 4. Electrode placement.

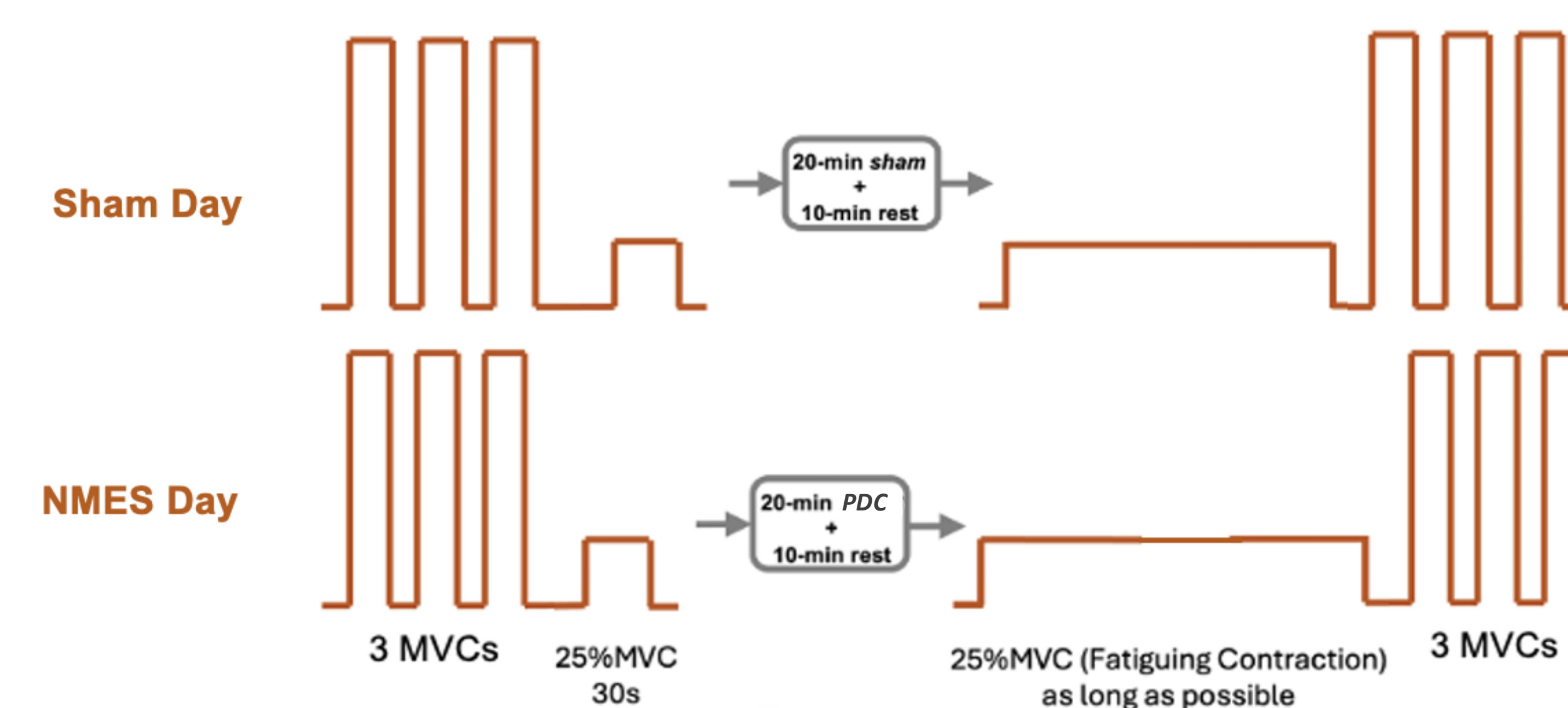


Figure 5. Experimental procedure.

Conclusion

- A single bout of neuromuscular electrical stimulation applied to the SS and IS significantly delayed neuromuscular fatigue, evidenced by longer endurance time and reduced median-frequency decline during sustained shoulder abduction.
- These findings suggest that pulsed current enhances neuromuscular endurance in individuals with RCD and may be an effective adjunct to rehabilitation programs aimed at improving fatigue resistance and shoulder function.

Reference

- [1] Thompson AK, Stein RB. Exp Brain Res. 2004;159(4):491–500.
- [2] Everaert DG, Thompson AK, Chong SL, Stein RB. Neurorehabil Neural Repair. 2010;24(2):168–177.
- [3] Taylor JL, Amann M, Duchateau J, Meeusen R, Rice CL. Med Sci Sports Exerc. 2016;48(11):2294–2306.